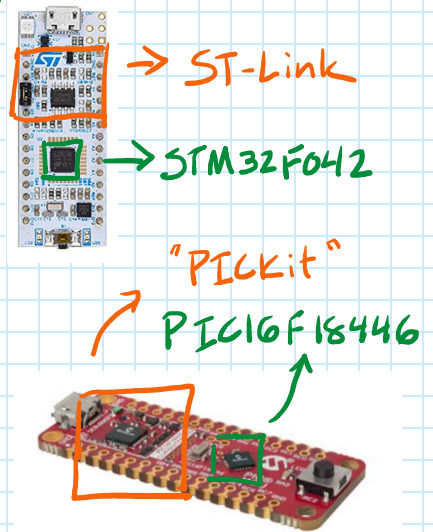
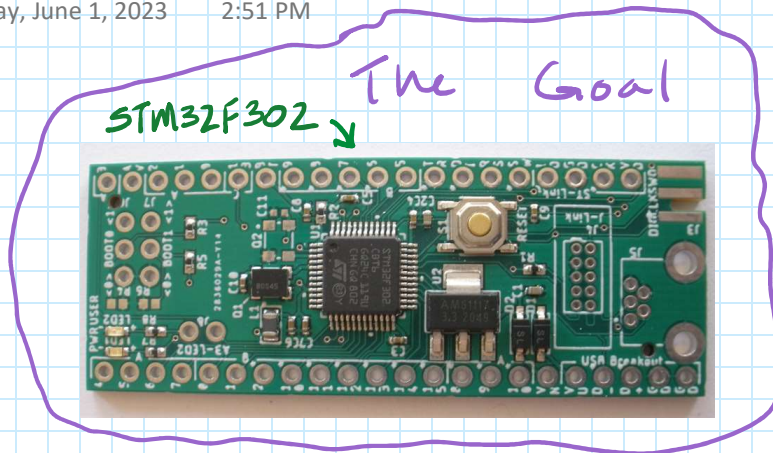
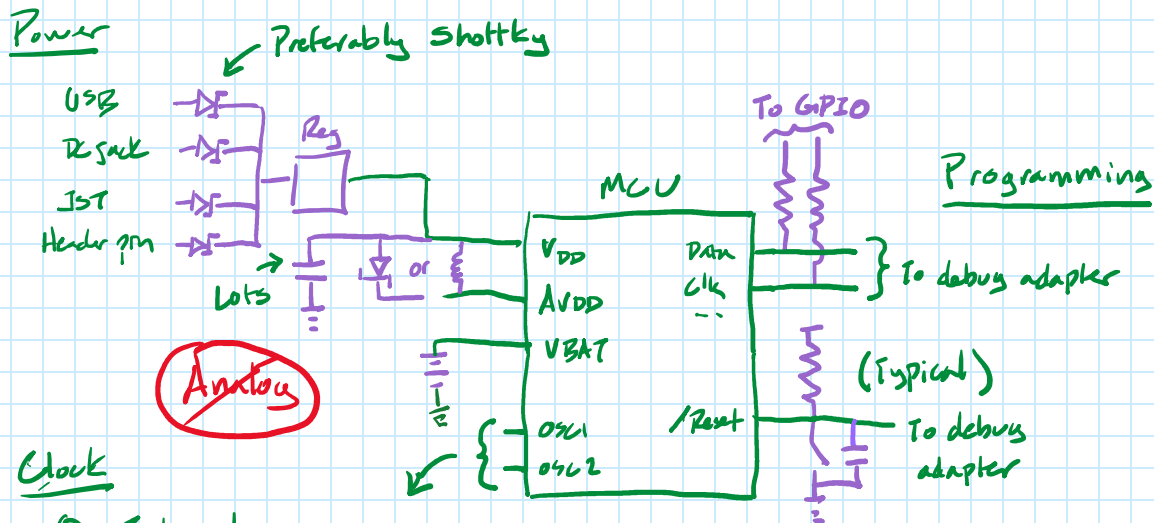


Make Your Own MCU Boards

Thursday, June 1, 2023 2:51 PM



① Draw the circuit → Check the MCU pinout & Hardware development guide



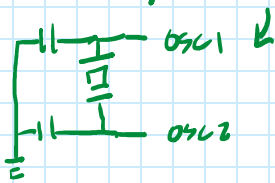
Clock

① Internal

↳ ~2-5% / Sometimes trimmable

② Oscillator

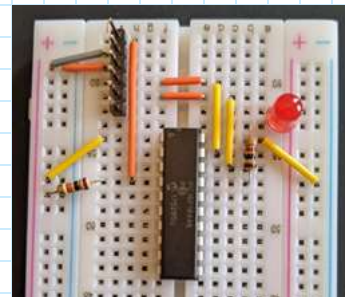
~~Crystal / Resonator~~



Misc

- Boot pins
- Unused pins / low power
- Power LED, LED_{BUILTIN}, BTNA

Ex bare-bones circuit for PIC16:



Make Your Own MCU Boards

Thursday, June 1, 2023 2:51 PM

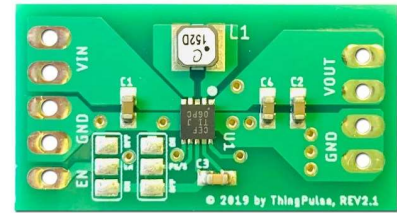
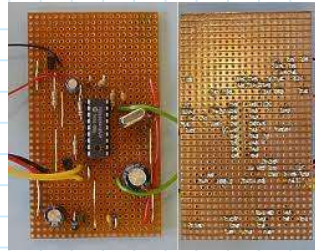
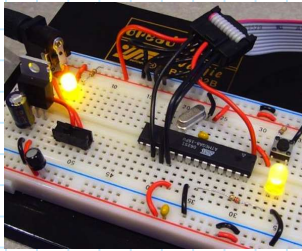
Don't forget the debug adapter!

ST-link
v2

V3

MPLAB
Snap

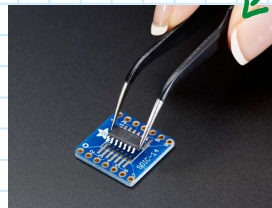
② Choose parts



Through-hole/Breakout

whatever you
can solder!

PCBA parts
library

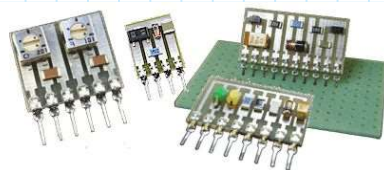


Large-ish SMT (0805,
1206, etc)



jlcpcb.com/parts

Adafruit #1210



capitaladvanced.com/6000ser.htm



Adafruit #4090



Digi-Key X209-ND



Adafruit #373

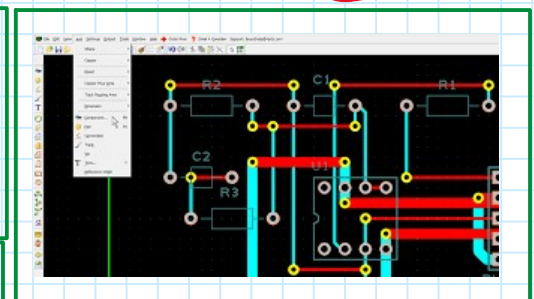
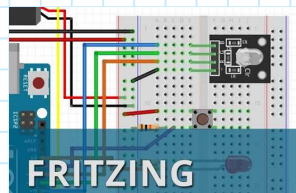
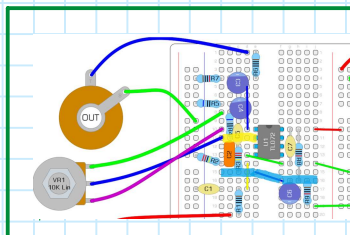
Electromechanical

- Switches - Switches, slide, toggle, buttons and jumpers/shunts
- Header - just headers!
- Terminal blocks
- Sockets - IC sockets (not female header)
- Other connectors - usb/communication, card holders
- Power connectors - battery holders, jacks & plugs
- Wire & Cables
- Standoffs & Spacers

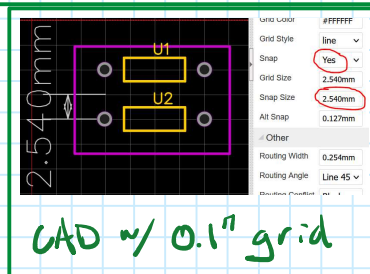
Passives

ladyada.net/wiki/partselector

③ Layout the circuit



EasyEDA →

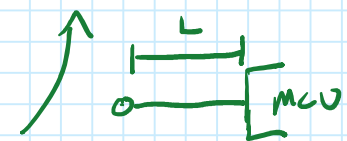


CAD ~ 0.1" grid

Max Freq?
~ 1s of MHz

① > 50 MHz

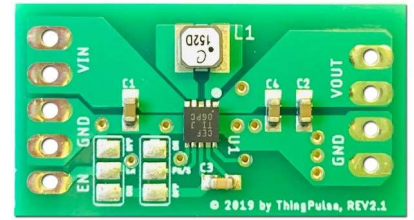
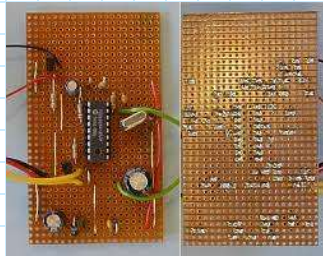
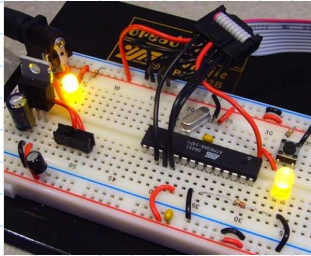
② > $\frac{6.1}{0.012 \cdot L}$ MHz *



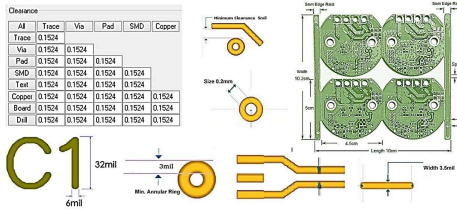
Make Your Own MCU Boards

Friday, June 9, 2023 7:41 AM

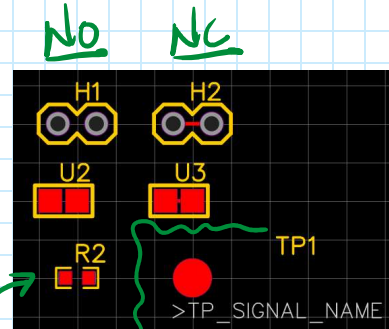
③ Layout the circuit (cont)



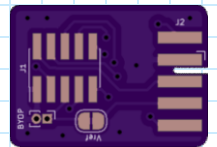
Basic PCB Design Rules



github.com/JLPCB/jlpcb-eagle
jlpcb.com/capabilities/Capabilities



Populate w/
OR to close

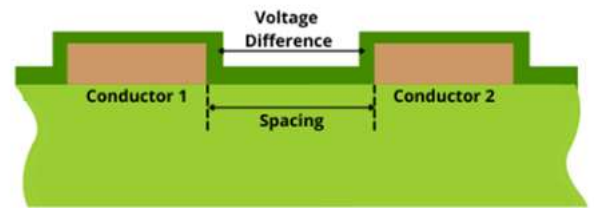


debug-edge.io

Max I? V?

10 mil trace, exterior = 3.5A

Closest traces 24 mil = 100 VDC
(headers @ 0.1" spacing)



www.protoexpress.com

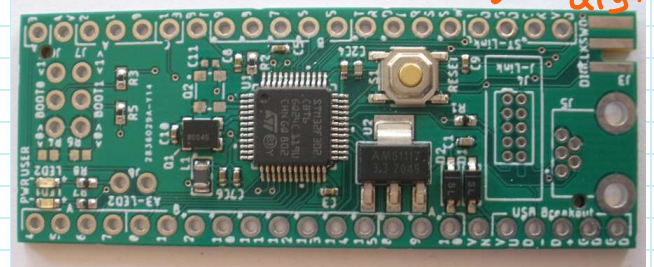
④ Construction



JLC PCB Assembly

$$\begin{aligned} \$ = & 9.5 + 0.0017 \cdot \text{SMT} + [3/\text{ext part}] \\ & + [3.5 + 0.0172 \cdot \text{PTH}] \end{aligned}$$

↖ \$3-5 e Qty: 10



Maker Fabs

$$\$ = 28 + 0.03 \cdot \text{SMT} + 0.04 \cdot \text{PTH}$$